SOLVABILITY

Exactly one solution

When a solution exists, it is unique if and only if the rank equals the number of unknowns.

UNIQUENESS

$$\operatorname{rank}(A) = n$$

HOLDS WHEN a solution exists, and n is the number of unknowns

✓ Unique

Full column rank

Every column has a pivot, no free variables, null space $\{\mathbf{0}\}$: $\mathbf{x}_p$ stands alone.

BOUNDARY

✗ Infinitely many

$\operatorname{rank}(A) < n$

$n - \operatorname{rank}(A)$ free variables, each a direction the solution can move along: a translated copy of the null space.

OUTSIDE THE TEST

existence $x_1 = 1, x_1 = 2$ has $\operatorname{rank}(A) = 1 = n$, yet $\operatorname{rank}([A \mid \mathbf{b}]) = 2$ and no solution. Uniqueness can be claimed only after the existence check.

b The rank test compares pivot columns of A with the unknowns; $\mathbf{b}$ plays no part, so the count is the same for every right-hand side.